

# Theory of Commutative Algebra 6 Theory Notes

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This file is a professionally reorganized and cleaned version.

# Contents

# Chapter 1

## Examples

### 1.1 Examples and intuition for $D(A)$

Recall that

$$D(A) = \{\mathfrak{p} \in \text{Spec}(A) : \exists a \in A \text{ such that } \mathfrak{p} \text{ is minimal in } V(\text{Ann}(a))\}.$$

Thus  $D(A)$  is built from annihilators of individual elements. It records the prime ideals that appear as the smallest prime-level reasons why some element is killed by multiplication. Equivalently, it is a prime-theoretic description of the set of zero divisors.

A useful slogan is:

$$\text{zero divisors} = \bigcup_{\mathfrak{p} \in D(A)} \mathfrak{p}.$$

#### 1.1.1 1. Domain case: $A = k[t]$

Let  $A = k[t]$ , where  $k$  is a field. Since  $A$  is an integral domain, the only zero divisor is 0. For every nonzero  $a \in A$ ,

$$\text{Ann}(a) = 0.$$

Hence the only prime that can arise is

$$D(A) = \{(0)\}.$$

Thus part (a) becomes

$$x \text{ is a zero divisor} \iff x \in (0),$$

which says exactly that only 0 is a zero divisor in a domain.

#### 1.1.2 2. Two crossing components: $A = k[x, y]/(xy)$

Let

$$A = k[x, y]/(xy),$$

and write  $\bar{x}, \bar{y}$  for the classes of  $x, y$  in  $A$ . Since

$$\bar{x}\bar{y} = 0,$$

both  $\bar{x}$  and  $\bar{y}$  are zero divisors.

Moreover,

$$\text{Ann}(\bar{x}) = (\bar{y}), \quad \text{Ann}(\bar{y}) = (\bar{x}).$$

Therefore

$$(\bar{x}), (\bar{y}) \in D(A).$$

In fact these are exactly the minimal primes of  $A$ , so

$$D(A) = \{(\bar{x}), (\bar{y})\}.$$

Hence the set of zero divisors is

$$(\bar{x}) \cup (\bar{y}).$$

**Geometric interpretation.** The scheme  $\text{Spec}(A)$  is the union of the two coordinate axes crossing at the origin. The primes in  $D(A)$  correspond to the irreducible components. An element is a zero divisor precisely when it vanishes on at least one whole component.

### 1.1.3 3. Nilpotent example: $A = k[t]/(t^2)$

Let

$$A = k[t]/(t^2),$$

and let  $\bar{t}$  denote the class of  $t$ . Then

$$\bar{t}^2 = 0,$$

so  $\bar{t}$  is nilpotent and hence a zero divisor.

Also

$$\text{Ann}(\bar{t}) = (\bar{t}).$$

Therefore

$$D(A) = \{(\bar{t})\}.$$

This example shows that  $D(A)$  detects zero divisors coming from nilpotent structure, not only from reducible geometry.

**Interpretation.** Here  $\text{Spec}(A)$  has only one point, but the ring is nonreduced. So  $D(A)$  can detect infinitesimal thickening as well as decomposition into components.

### 1.1.4 4. Localization removes primes that meet $S$

Again let

$$A = k[x, y]/(xy), \quad S = \{1, \bar{x}, \bar{x}^2, \dots\}.$$

In  $S^{-1}A$ , the element  $\bar{x}$  becomes invertible. Since

$$\bar{x}\bar{y} = 0,$$

it follows that

$$\bar{y} = 0$$

in  $S^{-1}A$ . Thus

$$S^{-1}A \cong k[\bar{x}, \bar{x}^{-1}],$$

which is a domain.

Hence

$$D(S^{-1}A) = \{(0)\}$$

inside  $\text{Spec}(S^{-1}A)$ .

Under the canonical identification

$$\text{Spec}(S^{-1}A) \cong \{\mathfrak{p} \in \text{Spec}(A) \mid \mathfrak{p} \cap S = \emptyset\},$$

this prime corresponds to  $(\bar{y})$ . Therefore, viewed as a subset of  $\text{Spec}(A)$ ,

$$D(S^{-1}A) = \{(\bar{y})\}.$$

But

$$D(A) = \{(\bar{x}), (\bar{y})\},$$

and only  $(\bar{y})$  is disjoint from  $S$ . Hence

$$D(S^{-1}A) = D(A) \cap \text{Spec}(S^{-1}A).$$

**Interpretation.** Localization keeps only the primes disjoint from  $S$ . Any prime in  $D(A)$  that meets  $S$  disappears after localization.

### 1.1.5 5. Embedded associated prime: $A = k[x, y]/(x^2, xy)$

Let

$$R = k[x, y], \quad I = (x^2, xy), \quad A = R/I.$$

Since

$$I = x(x, y) = (x) \cap (x^2, y),$$

we obtain a primary decomposition of  $I$ . Therefore the associated primes of  $A$  are

$$\text{Ass}(A) = \{(x)/I, (x, y)/I\}.$$

Now let us see that both belong to  $D(A)$ .

First,  $(x)/I$  is minimal over

$$\text{Ann}_A(1) = 0,$$

so

$$(x)/I \in D(A).$$

Second, the class  $\bar{x} \in A$  satisfies

$$x^2 = 0, \quad xy = 0.$$

Hence

$$\text{Ann}_A(\bar{x}) = (\bar{x}, \bar{y}) = (x, y)/I,$$

so

$$(x, y)/I \in D(A).$$

Thus

$$D(A) = \{(x)/I, (x, y)/I\} = \text{Ass}(A).$$

**Why this matters.** The prime  $(x, y)/I$  is an embedded prime, not a minimal prime. So  $D(A)$  can detect hidden nonreduced structure inside a component.

### 1.1.6 6. Product ring example: $A = k \times k$

Let

$$A = k \times k.$$

Then

$$(1, 0) \cdot (0, 1) = 0,$$

so both  $(1, 0)$  and  $(0, 1)$  are zero divisors.

The prime ideals are

$$\mathfrak{p}_1 = 0 \times k, \quad \mathfrak{p}_2 = k \times 0.$$

Also

$$\text{Ann}(1, 0) = 0 \times k = \mathfrak{p}_1, \quad \text{Ann}(0, 1) = k \times 0 = \mathfrak{p}_2.$$

Therefore

$$D(A) = \{\mathfrak{p}_1, \mathfrak{p}_2\}.$$

An element  $(a, b) \in k \times k$  is a zero divisor if and only if

$$a = 0 \quad \text{or} \quad b = 0.$$

So again

$$\{\text{zero divisors}\} = \mathfrak{p}_1 \cup \mathfrak{p}_2.$$

This is the most concrete algebraic model of a reducible space with two disjoint components.

### 1.1.7 7. Many zero divisors, one prime

Let

$$A = k[x, y]/(x^2, xy, y^2).$$

Then the maximal ideal

$$\mathfrak{m} = (\bar{x}, \bar{y})$$

satisfies

$$\mathfrak{m}^2 = 0.$$

Hence every element of  $\mathfrak{m}$  is nilpotent and therefore a zero divisor.

However,  $\text{Spec}(A)$  has only one prime ideal, namely  $\mathfrak{m}$ . Thus

$$D(A) = \{\mathfrak{m}\}.$$

This example shows that many different zero divisors may all be controlled by the same prime in  $D(A)$ .

### 1.1.8 8. Relation with $\text{Ass}(A)$

If the zero ideal of  $A$  has a primary decomposition, then part (c) says that

$$D(A) = \text{Ass}(A).$$

So in Noetherian rings with primary decomposition,  $D(A)$  is exactly the set of associated primes of the  $A$ -module  $A$ .

In particular:

- in reduced rings such as  $k[x, y]/(xy)$ ,  $D(A)$  often consists of minimal primes;
- in nonreduced rings such as  $k[x, y]/(x^2, xy)$ ,  $D(A)$  may also contain embedded primes.

### 1.1.9 9. Relation with zero divisors

Part (a) gives the important identity

$$\{\text{zero divisors of } A\} = \bigcup_{\mathfrak{p} \in D(A)} \mathfrak{p}.$$

Thus  $D(A)$  is a prime-theoretic description of the zero divisors. When primary decomposition exists, this becomes

$$\{\text{zero divisors of } A\} = \bigcup_{\mathfrak{p} \in \text{Ass}(A)} \mathfrak{p}.$$

### 1.1.10 10. How to compute $D(A)$ in practice

There are two standard methods.

**Method 1: via annihilators of elements.** Choose simple elements  $a \in A$ , compute  $\text{Ann}(a)$ , and find the minimal primes over  $\text{Ann}(a)$ . All primes obtained this way belong to  $D(A)$ .

**Method 2: via primary decomposition.** If  $A$  is Noetherian and  $0$  has a primary decomposition, compute that decomposition and take the radicals of its primary components. These radicals are exactly the primes in  $D(A)$ , because then

$$D(A) = \text{Ass}(A).$$

### 1.1.11 11. Final summary

The most important examples to remember are:

$$k[t], \quad k[x, y]/(xy), \quad k[t]/(t^2), \quad k[x, y]/(x^2, xy), \quad k \times k.$$

They illustrate, respectively:



- the domain case;
- reducible geometry with two components;
- nilpotent structure;
- embedded associated primes;
- and a completely explicit decomposed ring.

Together they show that  $D(A)$  measures where zero divisors come from, how those primes behave under localization, and why in the Noetherian case  $D(A)$  agrees with the set of associated primes.

## 1.2 Concrete polynomial and ring examples for $D(A)$

Recall that

$$D(A) = \{\mathfrak{p} \in \operatorname{Spec}(A) : \exists a \in A \text{ such that } \mathfrak{p} \text{ is minimal in } V(\operatorname{Ann}(a))\}.$$

Part (a) says that

$$\{\text{zero divisors of } A\} = \bigcup_{\mathfrak{p} \in D(A)} \mathfrak{p}.$$

We now present this on explicit rings and explicit polynomial quotients.

### 1.2.1 1. Arithmetic example: $A = \mathbb{Z}/6\mathbb{Z}$

The elements are

$$\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}.$$

The zero divisors are

$$\bar{0}, \bar{2}, \bar{3}, \bar{4},$$

because

$$\bar{2} \cdot \bar{3} = \bar{0}, \quad \bar{3} \cdot \bar{4} = \bar{0}.$$

Now compute annihilators:

$$\operatorname{Ann}(\bar{2}) = \{\bar{r} \in \mathbb{Z}/6\mathbb{Z} : \bar{r}\bar{2} = \bar{0}\} = \{\bar{0}, \bar{3}\} = (\bar{3}),$$

and

$$\operatorname{Ann}(\bar{3}) = \{\bar{0}, \bar{2}, \bar{4}\} = (\bar{2}).$$

The prime ideals of  $\mathbb{Z}/6\mathbb{Z}$  are

$$(\bar{2}), \quad (\bar{3}),$$

so

$$D(A) = \{(\bar{2}), (\bar{3})\}.$$

Thus

$$\{\text{zero divisors}\} = (\bar{2}) \cup (\bar{3}).$$

### 1.2.2 2. One-variable polynomial example: $A = k[x]/(x(x-1))$

Let  $\bar{x}$  denote the class of  $x$  in  $A$ . Then

$$\bar{x}(\bar{x} - 1) = 0,$$

so both  $\bar{x}$  and  $\bar{x} - 1$  are zero divisors.

We now compute their annihilators.

Since  $(\bar{x} - 1)\bar{x} = 0$ , every multiple of  $\bar{x} - 1$  annihilates  $\bar{x}$ . In fact,

$$\text{Ann}(\bar{x}) = (\bar{x} - 1).$$

Similarly,

$$\text{Ann}(\bar{x} - 1) = (\bar{x}).$$

Hence

$$(\bar{x}), (\bar{x} - 1) \in D(A),$$

and in fact

$$D(A) = \{(\bar{x}), (\bar{x} - 1)\}.$$

So the zero divisors are precisely the elements lying in one of these two prime ideals.

**Examples.**

$\bar{x}$  is a zero divisor,  $\bar{x} - 1$  is a zero divisor,  $\bar{1}$  is not a zero divisor.

**Geometric meaning.** The equation

$$x(x-1) = 0$$

describes the union of the two points  $x = 0$  and  $x = 1$  on the affine line.

### 1.2.3 3. Two-variable geometric example: $A = k[x, y]/(xy)$

Let  $\bar{x}, \bar{y}$  be the classes of  $x, y$  in  $A$ . Then

$$\bar{x}\bar{y} = 0,$$

so  $\bar{x}$  and  $\bar{y}$  are zero divisors.

To compute  $\text{Ann}(\bar{x})$ , we look for all  $\bar{f} \in A$  such that

$$\bar{f}\bar{x} = 0.$$

This means

$$fx \in (xy)$$

in  $k[x, y]$ . Since  $fx$  must be divisible by  $xy$ , the polynomial  $f$  must be divisible by  $y$ . Therefore

$$\text{Ann}(\bar{x}) = (\bar{y}).$$

Similarly,

$$\text{Ann}(\bar{y}) = (\bar{x}).$$

Hence

$$D(A) = \{(\bar{x}), (\bar{y})\}.$$

So the zero divisors are exactly the elements in

$$(\bar{x}) \cup (\bar{y}).$$

**Concrete classes.** Take

$$a = \bar{x}, \quad b = \bar{y}, \quad c = \bar{x} + \bar{y}, \quad d = \bar{x} + 1.$$

Then:

- $a$  is a zero divisor since  $a\bar{y} = 0$ ;
- $b$  is a zero divisor since  $b\bar{x} = 0$ ;
- $c \notin (\bar{x}) \cup (\bar{y})$ , so  $c$  is not a zero divisor;
- $d \notin (\bar{x}) \cup (\bar{y})$ , so  $d$  is not a zero divisor.

**Geometric meaning.** The variety

$$V(xy) = V(x) \cup V(y)$$

is the union of the two coordinate axes. The primes in  $D(A)$  correspond to those two components.

### 1.2.4 4. Nilpotent polynomial example: $A = k[t]/(t^2)$

Let  $\bar{t}$  be the class of  $t$  in  $A$ . Then

$$\bar{t}^2 = 0,$$

so  $\bar{t}$  is nilpotent and hence a zero divisor.

Now  $\bar{f} \in \text{Ann}(\bar{t})$  if and only if

$$\bar{f}\bar{t} = 0,$$

that is, if and only if

$$ft \in (t^2).$$

This holds exactly when  $f \in (t)$ . Hence

$$\text{Ann}(\bar{t}) = (\bar{t}),$$

and therefore

$$D(A) = \{(\bar{t})\}.$$

Every element of  $A$  has the form

$$a + b\bar{t}, \quad a, b \in k.$$

If  $a \neq 0$ , then  $a + b\bar{t}$  is a unit. If  $a = 0$ , then the element equals  $b\bar{t}$ , which is nilpotent. Thus the zero divisors are exactly the ideal

$$(\bar{t}).$$

### 1.2.5 5. Embedded-prime example: $A = k[x, y]/(x^2, xy)$

Let

$$A = k[x, y]/(x^2, xy),$$

and let  $\bar{x}, \bar{y}$  be the classes of  $x, y$ . Then in  $A$ ,

$$\bar{x}^2 = 0, \quad \bar{x}\bar{y} = 0.$$

So  $\bar{x}$  is annihilated by both  $\bar{x}$  and  $\bar{y}$ , and therefore

$$\text{Ann}(\bar{x}) = (\bar{x}, \bar{y}).$$

Hence

$$(\bar{x}, \bar{y}) \in D(A).$$

On the other hand,

$$(x^2, xy) = x(x, y) = (x) \cap (x^2, y),$$

so the associated primes are

$$(\bar{x}), \quad (\bar{x}, \bar{y}).$$

Therefore

$$D(A) = \{(\bar{x}), (\bar{x}, \bar{y})\}.$$

**Interpretation.** The prime  $(\bar{x})$  is the minimal prime, while  $(\bar{x}, \bar{y})$  is an embedded prime. This shows that  $D(A)$  can detect not only visible components but also hidden nilpotent structure inside a component.

### 1.2.6 6. Concrete localization example

Take again

$$A = k[x, y]/(xy),$$

and let

$$S = \{1, \bar{x}, \bar{x}^2, \bar{x}^3, \dots\}.$$

In the localization  $S^{-1}A$ , the element  $\bar{x}$  becomes invertible. Since

$$\bar{x}\bar{y} = 0,$$

we may multiply by  $\bar{x}^{-1}$  and obtain

$$\bar{y} = 0.$$

Hence

$$S^{-1}A \cong k[\bar{x}, \bar{x}^{-1}],$$

which is a domain.

Before localization,

$$D(A) = \{(\bar{x}), (\bar{y})\}.$$

After localization, the prime  $(\bar{x})$  disappears because it meets  $S$ , while  $(\bar{y})$  survives. Thus

$$D(S^{-1}A) = \{(\bar{y})\}$$

when viewed inside  $\text{Spec}(A)$ .

This is the concrete meaning of the formula

$$D(S^{-1}A) = D(A) \cap \text{Spec}(S^{-1}A).$$

## 1.2.7 7. Practical computation rule

For a quotient ring

$$A = k[x_1, \dots, x_n]/I,$$

one can compute  $D(A)$  concretely by the following steps:

1. Find obvious zero divisors, that is, classes whose product is zero modulo  $I$ .
2. For a chosen element  $a \in A$ , compute  $\text{Ann}(a)$  by solving

$$fa \in I.$$

3. Find the minimal primes over  $\text{Ann}(a)$ .
4. Collect all primes that arise.

If 0 has a primary decomposition, then one may instead compute  $\text{Ass}(A)$ , because in that case

$$D(A) = \text{Ass}(A).$$

## 1.2.8 8. Final summary

The most concrete examples to remember are:

$$\mathbb{Z}/6\mathbb{Z}, \quad k[x]/(x(x-1)), \quad k[x, y]/(xy), \quad k[t]/(t^2), \quad k[x, y]/(x^2, xy).$$

They illustrate:

- arithmetic zero divisors;
- decomposition into two points;
- decomposition into two axes;
- nilpotent zero divisors;
- and embedded associated primes.

## 1.3 Classes in quotient rings, minimal primes, and associated primes

### 1.3.1 1. What do $\bar{x}, \bar{y}$ mean?

Let

$$A = R/I$$

be a quotient ring. For any element  $f \in R$ , its image in  $A$  is denoted by

$$\bar{f} \quad \text{or sometimes} \quad [f].$$

Thus

$$\bar{f} = f + I.$$

So:

$$\bar{x} = x + I, \quad \bar{y} = y + I, \quad \overline{x + y} = x + y + I.$$

This is the residue class (or coset) of  $f$  modulo  $I$ .

Two elements  $f, g \in R$  define the same class in  $A$  if and only if

$$\bar{f} = \bar{g} \iff f - g \in I.$$

### 1.3.2 2. Concrete examples of classes

**Example:**  $A = k[x, y]/(xy)$

Here  $I = (xy)$ . Then:

$$\bar{x} = x + (xy), \quad \bar{y} = y + (xy).$$

Also

$$\overline{x + xy} = \bar{x}$$

because

$$(x + xy) - x = xy \in (xy).$$

Moreover,

$$\overline{xy} = 0, \quad \overline{x^2y} = 0,$$

since  $xy, x^2y \in (xy)$ .

However,

$$\bar{x} \neq 0, \quad \bar{y} \neq 0,$$

because  $x, y \notin (xy)$ .

Thus in this quotient ring,

$$\bar{x}\bar{y} = 0$$

while neither  $\bar{x}$  nor  $\bar{y}$  is zero.

**Example:**  $A = k[t]/(t^2)$

Here

$$\bar{t} \neq 0$$

but

$$\bar{t}^2 = 0.$$

Every element has the form

$$a + b\bar{t}, \quad a, b \in k.$$

**Example:**  $A = \mathbb{Z}/6\mathbb{Z}$

The class  $\bar{2}$  means all integers congruent to 2 modulo 6:

$$\bar{2} = \{\dots, -10, -4, 2, 8, 14, \dots\}.$$

Also

$$\bar{2}\bar{3} = \bar{0}.$$

### 1.3.3 3. How to think about quotient classes

The quotient ring  $R/I$  is obtained from  $R$  by imposing all relations coming from  $I$ .

For example:

$k[x, y]/(xy)$  means that in the quotient we force  $xy = 0$ ;

$k[x, y]/(x^2, xy)$  means that in the quotient we force  $x^2 = 0$  and  $xy = 0$ ;

$k[x]/(x(x-1))$  means that in the quotient we force  $x(x-1) = 0$ .

### 1.3.4 4. Minimal primes

A prime ideal  $\mathfrak{p} \subset A$  is called a *minimal prime* if it is minimal among the prime ideals of  $A$ .

If

$$A = R/I,$$

then minimal primes of  $A$  correspond to prime ideals of  $R$  that are minimal over  $I$ .

We write

$$\text{Min}(A)$$

for the set of minimal primes.

**Interpretation.** Minimal primes describe the top-level irreducible components of the ring or affine scheme.

**Example:**  $A = k[x]$

Since  $A$  is a domain,  $(0)$  is prime and

$$\text{Min}(A) = \{(0)\}.$$

**Example:**  $A = k[x, y]/(xy)$

The ideal  $(xy)$  has two minimal primes:

$$(x), \quad (y).$$

Hence in the quotient,

$$\text{Min}(A) = \{(\bar{x}), (\bar{y})\}.$$

**Example:**  $A = k[x, y]/(x^2)$

The only prime minimal over  $(x^2)$  is  $(x)$ . Hence

$$\text{Min}(A) = \{(\bar{x})\}.$$

**Example:**  $A = k[x, y]/(x^2, xy)$

Since

$$\sqrt{(x^2, xy)} = (x),$$

the only minimal prime is

$$\text{Min}(A) = \{(\bar{x})\}.$$

### 1.3.5 5. Associated primes

A prime ideal  $\mathfrak{p} \subset A$  is an *associated prime* of the  $A$ -module  $A$  if

$$\mathfrak{p} = \text{Ann}_A(a)$$

for some element  $a \in A$ .

We write

$$\text{Ass}(A)$$

for the set of associated primes of  $A$ .

**Interpretation.** Associated primes describe primes that arise as actual annihilators of elements. They detect zero-divisor structure and possible embedded components.

**Example:**  $A = k[x, y]/(xy)$

We have

$$\text{Ann}(\bar{x}) = (\bar{y}), \quad \text{Ann}(\bar{y}) = (\bar{x}).$$

Therefore

$$\text{Ass}(A) = \{(\bar{x}), (\bar{y})\}.$$

Here

$$\text{Ass}(A) = \text{Min}(A).$$

**Example:**  $A = k[t]/(t^2)$

We have

$$\text{Ann}(\bar{t}) = (\bar{t}).$$

Hence

$$\text{Ass}(A) = \{(\bar{t})\} = \text{Min}(A).$$



**Example:**  $A = k[x, y]/(x^2, xy)$

First,

$$\text{Min}(A) = \{(\bar{x})\}.$$

Now in  $A$ ,

$$\bar{x}^2 = 0, \quad \bar{x}\bar{y} = 0.$$

Hence

$$(\bar{x}, \bar{y}) \subseteq \text{Ann}(\bar{x}).$$

Conversely, if  $f\bar{x} = 0$ , then

$$fx \in (x^2, xy) = x(x, y),$$

so  $f \in (x, y)$ . Therefore

$$\text{Ann}(\bar{x}) = (\bar{x}, \bar{y}).$$

Thus

$$(\bar{x}, \bar{y}) \in \text{Ass}(A).$$

Since  $(\bar{x})$  is the minimal prime, it is also associated in this Noetherian case. Hence

$$\text{Ass}(A) = \{(\bar{x}), (\bar{x}, \bar{y})\}.$$

So

$$\text{Min}(A) = \{(\bar{x})\} \subsetneq \text{Ass}(A) = \{(\bar{x}), (\bar{x}, \bar{y})\}.$$

The prime  $(\bar{x}, \bar{y})$  is an embedded associated prime.

### 1.3.6 6. Comparison

Minimal primes describe the irreducible components of the ring:

geometry at the top level.

Associated primes describe annihilators of actual elements:

zero-divisor and embedded structure.

Thus in general,

$$\text{Min}(A) \subseteq \text{Ass}(A),$$

and the inclusion may be strict.

### 1.3.7 7. Summary table

| $A$                 | $\text{Min}(A)$            | $\text{Ass}(A)$                     |
|---------------------|----------------------------|-------------------------------------|
| $k[x]$              | $\{(0)\}$                  | $\{(0)\}$                           |
| $k[x, y]/(xy)$      | $\{(\bar{x}), (\bar{y})\}$ | $\{(\bar{x}), (\bar{y})\}$          |
| $k[t]/(t^2)$        | $\{(\bar{t})\}$            | $\{(\bar{t})\}$                     |
| $k[x, y]/(x^2, xy)$ | $\{(\bar{x})\}$            | $\{(\bar{x}), (\bar{x}, \bar{y})\}$ |

### 1.3.8 8. Relation with $D(A)$

In the problem,  $D(A)$  is defined via primes minimal over annihilators of elements. In the Noetherian case with primary decomposition,

$$D(A) = \text{Ass}(A).$$

Moreover, the set of zero divisors is

$$\bigcup_{\mathfrak{p} \in D(A)} \mathfrak{p}.$$

So associated primes are the natural language for understanding zero divisors.

## 1.4 Concrete meaning of $x$ , $x^2$ , $\mathbb{Z}/p\mathbb{Z}$ , $\mathbb{Z}/n\mathbb{Z}$ , quotient classes, minimal primes, and associated primes

### 1.4.1 1. What is $x$ in a polynomial ring?

In a polynomial ring

$$R[x],$$

the symbol  $x$  is a *formal indeterminate*. It is not automatically a number. A polynomial is a finite expression

$$a_0 + a_1x + \cdots + a_nx^n \quad (a_i \in R).$$

Thus in  $\mathbb{Z}[x]$ ,

$$2x^2 + 3x + 5$$

is a polynomial with integer coefficients. Here  $x$  is only a formal symbol.

### 1.4.2 2. What is $x^2$ ?

The symbol

$$x^2$$

means  $x \cdot x$ , the square of the indeterminate  $x$ . It is itself a polynomial. For example:

$$x = 0 + 1 \cdot x, \quad x^2 = 0 + 0 \cdot x + 1 \cdot x^2.$$

### 1.4.3 3. Evaluation

Once one chooses an element  $a \in R$ , one may evaluate:

$$f(x) \mapsto f(a).$$

For example, if

$$f(x) = 2x^2 + 3x + 5 \in \mathbb{Z}[x],$$

then

$$f(4) = 2 \cdot 4^2 + 3 \cdot 4 + 5 = 49.$$

Thus  $x$  is formal in the polynomial ring, but becomes an actual value under evaluation.

**1.4.4 4. The ring  $\mathbb{Z}/n\mathbb{Z}$** 

The ring

$$\mathbb{Z}/n\mathbb{Z}$$

consists of residue classes modulo  $n$ :

$$\bar{0}, \bar{1}, \dots, \overline{n-1}.$$

Two integers define the same class if and only if they differ by a multiple of  $n$ :

$$a \equiv b \pmod{n} \iff n \mid (a - b).$$

Example in  $\mathbb{Z}/6\mathbb{Z}$ :

$$\bar{2} = \bar{8} = \overline{-4}.$$

Arithmetic is done modulo  $n$ .

**1.4.5 5. The ring  $\mathbb{Z}/p\mathbb{Z}$** 

If  $p$  is prime, then

$$\mathbb{Z}/p\mathbb{Z}$$

is a field, often denoted

$$\mathbb{F}_p.$$

For example, in  $\mathbb{Z}/5\mathbb{Z}$ ,

$$\bar{2} + \bar{4} = \bar{1}, \quad \bar{2}\bar{3} = \bar{1}.$$

If  $n$  is not prime, then  $\mathbb{Z}/n\mathbb{Z}$  usually has zero divisors. For example, in  $\mathbb{Z}/6\mathbb{Z}$ ,

$$\bar{2} \cdot \bar{3} = \bar{0}.$$

**1.4.6 6. Polynomial rings over  $\mathbb{Z}/p\mathbb{Z}$** 

The ring

$$(\mathbb{Z}/5\mathbb{Z})[x]$$

consists of polynomials in  $x$  with coefficients modulo 5. For example:

$$\bar{3}x^2 + \bar{4}x + \bar{1}.$$

Addition and multiplication are performed as usual, with coefficients reduced modulo 5.

**1.4.7 7. Quotient classes**

If

$$A = R/I,$$

then for  $f \in R$ , its class in  $A$  is

$$\bar{f} = f + I.$$

Thus:

$$\bar{x} = x + I, \quad \overline{x^2} = x^2 + I.$$

Two elements define the same class if and only if their difference lies in  $I$ :

$$\bar{f} = \bar{g} \iff f - g \in I.$$

**1.4.8 8. Example:  $k[x]/(x^2)$** 

Let

$$A = k[x]/(x^2).$$

Then:

$$\bar{x} \neq 0$$

because  $x \notin (x^2)$ , but

$$\bar{x}^2 = 0$$

because  $x^2 \in (x^2)$ .

Every element of  $A$  has the form

$$a + b\bar{x}, \quad a, b \in k.$$

**1.4.9 9. Example:  $k[x]/(x(x-1))$** 

Let

$$A = k[x]/(x(x-1)).$$

Then in  $A$ ,

$$\bar{x}(\bar{x} - 1) = 0.$$

So  $\bar{x}$  and  $\bar{x} - 1$  are zero divisors.

The minimal primes are

$$(\bar{x}), \quad (\bar{x} - 1).$$

Hence

$$\text{Min}(A) = \{(\bar{x}), (\bar{x} - 1)\}.$$

Also

$$\text{Ass}(A) = \{(\bar{x}), (\bar{x} - 1)\}.$$

**1.4.10 10. Example:**  $k[x, y]/(xy)$ 

Let

$$A = k[x, y]/(xy).$$

Then

$$\bar{x}\bar{y} = 0,$$

but  $\bar{x} \neq 0$  and  $\bar{y} \neq 0$ .

The minimal primes are

$$(\bar{x}), \quad (\bar{y}),$$

so

$$\text{Min}(A) = \{(\bar{x}), (\bar{y})\}.$$

Also

$$\text{Ann}(\bar{x}) = (\bar{y}), \quad \text{Ann}(\bar{y}) = (\bar{x}),$$

hence

$$\text{Ass}(A) = \{(\bar{x}), (\bar{y})\}.$$

**1.4.11 11. Example:**  $k[x, y]/(x^2, xy)$ 

Let

$$A = k[x, y]/(x^2, xy).$$

Then

$$\bar{x}^2 = 0, \quad \bar{x}\bar{y} = 0.$$

The radical of  $(x^2, xy)$  is  $(x)$ , so

$$\text{Min}(A) = \{(\bar{x})\}.$$

Now

$$\text{Ann}(\bar{x}) = (\bar{x}, \bar{y}),$$

so

$$(\bar{x}, \bar{y}) \in \text{Ass}(A).$$

Thus

$$\text{Ass}(A) = \{(\bar{x}), (\bar{x}, \bar{y})\}.$$

Therefore

$$\text{Min}(A) \subsetneq \text{Ass}(A).$$

The prime  $(\bar{x}, \bar{y})$  is an embedded associated prime.

**1.4.12 12. Interpretation**

Minimal primes describe the main irreducible components of the ring. Associated primes describe actual annihilators of elements, hence zero-divisor structure and possible embedded components.

Thus:

minimal primes = top geometry,      associated primes = top geometry + hidden annihilation structure

## 1.5 Concrete meaning of $x$ , $x^2$ , $\mathbb{Z}/n\mathbb{Z}$ , $\mathbb{Z}/p\mathbb{Z}$ , quotient classes, minimal primes, and associated primes

### 1.5.1 1. Polynomial rings and formal indeterminates

Let  $R$  be a ring. The polynomial ring

$$R[x]$$

consists of all finite expressions

$$a_0 + a_1x + \cdots + a_nx^n, \quad a_i \in R.$$

Here  $x$  is a *formal indeterminate*. It is not automatically a number. It is simply a symbol used to build polynomials.

Thus:

$$x = 0 + 1 \cdot x, \quad x^2 = 0 + 0 \cdot x + 1 \cdot x^2.$$

So  $x^2$  means the product  $x \cdot x$ , and it is itself a polynomial.

For example, in  $\mathbb{Z}[x]$ ,

$$2x^2 + 3x + 5$$

is a polynomial with integer coefficients. At this stage,  $x$  is not evaluated at any integer.

### 1.5.2 2. Evaluation of polynomials

If  $f(x) \in R[x]$  and  $a \in R$ , then we may evaluate  $f$  at  $a$ :

$$f(x) \mapsto f(a).$$

For example, if

$$f(x) = 2x^2 + 3x + 5 \in \mathbb{Z}[x],$$

then

$$f(4) = 2 \cdot 4^2 + 3 \cdot 4 + 5 = 32 + 12 + 5 = 49.$$

Thus there is a fundamental distinction:

- in  $R[x]$ ,  $x$  is formal;
- after evaluation,  $x$  is replaced by an actual element of a ring.

### 1.5.3 3. The rings $\mathbb{Z}/n\mathbb{Z}$ and $\mathbb{Z}/p\mathbb{Z}$

The ring

$$\mathbb{Z}/n\mathbb{Z}$$

is the ring of integers modulo  $n$ . Its elements are residue classes

$$\bar{0}, \bar{1}, \bar{2}, \dots, \overline{n-1},$$

where

$$\bar{a} = \bar{b} \iff a \equiv b \pmod{n} \iff n \mid (a - b).$$

**Example:**  $\mathbb{Z}/6\mathbb{Z}$

The classes are

$$\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}, \bar{5}.$$

We have

$$\bar{2} = \bar{8} = \overline{-4},$$

and arithmetic is done modulo 6:

$$\bar{4} + \bar{5} = \bar{9} = \bar{3}, \quad \bar{2} \cdot \bar{4} = \bar{8} = \bar{2}.$$

Also

$$\bar{2} \cdot \bar{3} = \bar{0},$$

so  $\bar{2}$  and  $\bar{3}$  are zero divisors.

**Example:**  $\mathbb{Z}/5\mathbb{Z}$

Since 5 is prime, the ring

$$\mathbb{Z}/5\mathbb{Z}$$

is a field, often denoted  $\mathbb{F}_5$ . Its elements are

$$\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}.$$

Examples:

$$\bar{2} + \bar{4} = \bar{6} = \bar{1}, \quad \bar{2} \cdot \bar{3} = \bar{6} = \bar{1}.$$

Every nonzero element has a multiplicative inverse.

In general:

- $\mathbb{Z}/p\mathbb{Z}$  is a field if  $p$  is prime;
- $\mathbb{Z}/n\mathbb{Z}$  usually has zero divisors if  $n$  is composite.

### 1.5.4 4. Polynomial rings over $\mathbb{Z}/p\mathbb{Z}$

One may form polynomial rings over finite fields:

$$(\mathbb{Z}/p\mathbb{Z})[x].$$

For example, in  $(\mathbb{Z}/5\mathbb{Z})[x]$ ,

$$f(x) = \bar{3}x^2 + \bar{4}x + \bar{1}$$

is a polynomial.

Addition is coefficientwise modulo 5. For example,

$$(\bar{3}x^2 + \bar{4}x + \bar{1}) + (\bar{4}x^2 + \bar{2}x + \bar{3}) = \bar{2}x^2 + \bar{x} + \bar{4},$$

because

$$\bar{3} + \bar{4} = \bar{2}, \quad \bar{4} + \bar{2} = \bar{1}, \quad \bar{1} + \bar{3} = \bar{4}.$$

### 1.5.5 5. Quotient rings and classes

Let  $R$  be a ring and  $I \subseteq R$  an ideal. The quotient ring

$$R/I$$

consists of residue classes modulo  $I$ . For  $f \in R$ , its class is written

$$\bar{f} = f + I.$$

Thus:

$$\bar{x} = x + I, \quad \overline{x^2} = x^2 + I.$$

Two elements  $f, g \in R$  define the same class if and only if

$$\bar{f} = \bar{g} \iff f - g \in I.$$

So  $\bar{f}$  is not the original polynomial  $f$ , but its equivalence class modulo  $I$ .

### 1.5.6 6. Example: $k[x]/(x^2)$

Let

$$A = k[x]/(x^2),$$

where  $k$  is a field.

In this quotient ring, the class  $\bar{x}$  satisfies

$$\bar{x}^2 = 0,$$

because  $x^2 \in (x^2)$ . However,

$$\bar{x} \neq 0,$$

because  $x \notin (x^2)$ .

Every element of  $A$  has the form

$$a + b\bar{x}, \quad a, b \in k.$$

Indeed, every polynomial

$$f(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$$

has class

$$\bar{f} = a_0 + a_1\bar{x},$$

since all terms of degree at least 2 vanish in the quotient.

**Examples.**

$$\overline{x + x^2} = \bar{x}, \quad \overline{1 + x^3} = 1, \quad (\bar{x})^2 = 0.$$

This is a standard example of a nonreduced ring:  $\bar{x} \neq 0$ , but  $\bar{x}$  is nilpotent.



**1.5.7 7. Example:**  $k[x]/(x(x-1))$ 

Let

$$A = k[x]/(x(x-1)).$$

Then in  $A$ ,

$$\bar{x}(\bar{x}-1) = 0.$$

Hence  $\bar{x}$  and  $\bar{x}-1$  are zero divisors.

This ring encodes the relation

$$x(x-1) = 0.$$

Geometrically, it corresponds to two points on the affine line:  $x = 0$  and  $x = 1$ .

**Examples.**

$$\bar{x} \neq 0, \quad \bar{x}-1 \neq 0, \quad \bar{x}(\bar{x}-1) = 0.$$

**1.5.8 8. Example:**  $k[x, y]/(xy)$ 

Let

$$A = k[x, y]/(xy).$$

Then

$$\bar{x}\bar{y} = 0,$$

but neither  $\bar{x}$  nor  $\bar{y}$  is zero. Indeed,

$$x \notin (xy), \quad y \notin (xy).$$

This is one of the most important examples:

$$\bar{x}\bar{y} = 0 \quad \text{with} \quad \bar{x} \neq 0, \quad \bar{y} \neq 0.$$

So  $\bar{x}$  and  $\bar{y}$  are nonzero zero divisors.

**Examples of classes.**

$$\overline{x+xy} = \bar{x}, \quad \overline{y+x^2y} = \bar{y}, \quad \overline{xy} = 0.$$

**Annihilators.** We compute:

$$\text{Ann}(\bar{x}) = (\bar{y}), \quad \text{Ann}(\bar{y}) = (\bar{x}).$$

Indeed,  $\bar{f} \in \text{Ann}(\bar{x})$  means

$$\bar{f}\bar{x} = 0 \iff fx \in (xy),$$

which forces  $f \in (y)$ . Similarly for  $\text{Ann}(\bar{y})$ .

**1.5.9 9. Example:**  $k[x, y]/(x^2, xy)$ 

Let

$$A = k[x, y]/(x^2, xy).$$

Then in  $A$ ,

$$\bar{x}^2 = 0, \quad \bar{x}\bar{y} = 0.$$

Thus  $\bar{x}$  is killed by both  $\bar{x}$  and  $\bar{y}$ .

We compute:

$$\text{Ann}(\bar{x}) = (\bar{x}, \bar{y}).$$

Indeed, if  $f\bar{x} = 0$ , then

$$fx \in (x^2, xy) = x(x, y),$$

so  $f \in (x, y)$ .

This example is very important because it has an embedded associated prime.

**1.5.10 10. Minimal primes**

Let  $A = R/I$ . A prime ideal  $\mathfrak{p} \subseteq R$  is *minimal over  $I$*  if:

- $I \subseteq \mathfrak{p}$ ,
- $\mathfrak{p}$  is prime,
- there is no smaller prime ideal between  $I$  and  $\mathfrak{p}$ .

The corresponding primes in  $A = R/I$  are called the *minimal primes* of  $A$ . We write

$$\text{Min}(A)$$

for the set of minimal primes.

**Interpretation.** Minimal primes describe the top-level irreducible components of the ring or affine variety.

**Example:**  $k[x]$

Since  $k[x]$  is a domain, the zero ideal is prime. Thus

$$\text{Min}(k[x]) = \{(0)\}.$$

**Example:**  $k[x]/(x^2)$

The radical of  $(x^2)$  is  $(x)$ , so

$$\text{Min}(k[x]/(x^2)) = \{(\bar{x})\}.$$

**Example:**  $k[x]/(x(x-1))$

The ideal  $(x(x-1))$  has two minimal primes:

$$(x), \quad (x-1).$$

Hence

$$\text{Min}\left(k[x]/(x(x-1))\right) = \{(\bar{x}), (\bar{x}-1)\}.$$

**Example:**  $k[x, y]/(xy)$

The ideal  $(xy)$  has two minimal primes:

$$(x), \quad (y).$$

Therefore

$$\text{Min}\left(k[x, y]/(xy)\right) = \{(\bar{x}), (\bar{y})\}.$$

**Example:**  $k[x, y]/(x^2, xy)$

We have

$$\sqrt{(x^2, xy)} = (x),$$

so

$$\text{Min}\left(k[x, y]/(x^2, xy)\right) = \{(\bar{x})\}.$$

### 1.5.11 11. Associated primes

A prime ideal  $\mathfrak{p} \subseteq A$  is an *associated prime* of the  $A$ -module  $A$  if

$$\mathfrak{p} = \text{Ann}_A(a)$$

for some  $a \in A$ .

We write

$$\text{Ass}(A)$$

for the set of associated primes.

**Interpretation.** Associated primes detect actual annihilators of elements. They describe zero-divisor structure and may reveal hidden embedded components.

**Example:**  $k[x]/(x^2)$

We have

$$\text{Ann}(\bar{x}) = (\bar{x}),$$

so

$$\text{Ass}\left(k[x]/(x^2)\right) = \{(\bar{x})\}.$$

**Example:**  $k[x]/(x(x-1))$

We have

$$\text{Ann}(\bar{x}) = (\bar{x} - 1), \quad \text{Ann}(\bar{x} - 1) = (\bar{x}).$$

Hence

$$\text{Ass}\left(k[x]/(x(x-1))\right) = \{(\bar{x}), (\bar{x} - 1)\}.$$

**Example:**  $k[x, y]/(xy)$

We computed

$$\text{Ann}(\bar{x}) = (\bar{y}), \quad \text{Ann}(\bar{y}) = (\bar{x}),$$

so

$$\text{Ass}\left(k[x, y]/(xy)\right) = \{(\bar{x}), (\bar{y})\}.$$

**Example:**  $k[x, y]/(x^2, xy)$

We have already seen that

$$\text{Ann}(\bar{x}) = (\bar{x}, \bar{y}).$$

Also the minimal prime  $(\bar{x})$  is associated in this Noetherian case. Therefore

$$\text{Ass}\left(k[x, y]/(x^2, xy)\right) = \{(\bar{x}), (\bar{x}, \bar{y})\}.$$

Thus

$$\text{Min}(A) = \{(\bar{x})\} \subsetneq \text{Ass}(A) = \{(\bar{x}), (\bar{x}, \bar{y})\}.$$

The prime  $(\bar{x}, \bar{y})$  is an *embedded associated prime*.

### 1.5.12 12. Comparison of minimal primes and associated primes

Minimal primes describe the irreducible components:

top-level geometry.

Associated primes describe annihilators of actual elements:

zero-divisor structure and embedded behavior.

In general, for Noetherian rings,

$$\text{Min}(A) \subseteq \text{Ass}(A),$$

and the inclusion may be strict.

### 1.5.13 13. Summary of the main examples

| $A$                 | $\text{Min}(A)$              | $\text{Ass}(A)$                     |
|---------------------|------------------------------|-------------------------------------|
| $k[x]$              | $\{(0)\}$                    | $\{(0)\}$                           |
| $k[x]/(x^2)$        | $\{(\bar{x})\}$              | $\{(\bar{x})\}$                     |
| $k[x]/(x(x-1))$     | $\{(\bar{x}), (\bar{x}-1)\}$ | $\{(\bar{x}), (\bar{x}-1)\}$        |
| $k[x, y]/(xy)$      | $\{(\bar{x}), (\bar{y})\}$   | $\{(\bar{x}), (\bar{y})\}$          |
| $k[x, y]/(x^2, xy)$ | $\{(\bar{x})\}$              | $\{(\bar{x}), (\bar{x}, \bar{y})\}$ |

### 1.5.14 14. Final intuition

The key points are:

- $x, y$  are formal variables in polynomial rings;
- $x^2, xy$  are formal products;
- quotient rings impose algebraic relations such as  $x^2 = 0$  or  $xy = 0$ ;
- $\bar{x}, \bar{y}$  are classes of  $x, y$  in the quotient;
- minimal primes describe the main geometric components;
- associated primes describe where actual annihilation occurs.

Thus one may think of:

minimal primes = geometry,      associated primes = geometry plus hidden zero-divisor structure.

## 1.6 Concrete annihilators: $\mathbb{Z}/p\mathbb{Z}$ , $\mathbb{Z}/n\mathbb{Z}$ , polynomial rings, and quotient rings

### 1.6.1 1. Definition of annihilator

Let  $A$  be a ring and let  $a \in A$ . The annihilator of  $a$  is

$$\text{Ann}_A(a) = \{r \in A : ra = 0\}.$$

It is an ideal of  $A$ .

More generally, if  $S \subseteq A$ , then

$$\text{Ann}_A(S) = \{r \in A : rs = 0 \text{ for all } s \in S\}.$$

For a single element  $a$ , this is just  $\text{Ann}_A(a)$ .

**Basic intuition.** The annihilator of  $a$  measures which elements of the ring kill  $a$  by multiplication. So:

- if  $\text{Ann}_A(a) = 0$ , then no nonzero element kills  $a$ ;
- if  $\text{Ann}_A(a) \neq 0$ , then  $a$  is a zero divisor (unless  $a = 0$ );
- prime annihilators are exactly what lead to associated primes.

### 1.6.2 2. First trivial examples

For every ring  $A$ ,

$$\text{Ann}_A(0) = A,$$

because every element kills 0.

Also,

$$\text{Ann}_A(1) = 0,$$

because if  $r \cdot 1 = 0$ , then  $r = 0$ .

Thus in every nonzero ring:

$$\text{Ann}_A(0) = A, \quad \text{Ann}_A(1) = 0.$$

### 1.6.3 3. Annihilators in a domain

Suppose  $A$  is an integral domain. Then for every nonzero  $a \in A$ ,

$$\text{Ann}_A(a) = 0.$$

Indeed, if  $ra = 0$  and  $a \neq 0$ , then  $r = 0$  because there are no zero divisors.

**Examples.** In  $k[x]$ , where  $k$  is a field,

$$\text{Ann}_{k[x]}(x) = 0, \quad \text{Ann}_{k[x]}(x^2 + 1) = 0, \quad \text{Ann}_{k[x]}(f) = 0$$

for every nonzero polynomial  $f \in k[x]$ .

So in a domain, annihilators of nonzero elements are always trivial.

### 1.6.4 4. Annihilators in the field $\mathbb{Z}/p\mathbb{Z}$

Let  $p$  be prime. Then

$$\mathbb{Z}/p\mathbb{Z}$$

is a field. Therefore every nonzero element is invertible, hence cannot have a nonzero annihilator.

So:

$$\text{Ann}_{\mathbb{Z}/p\mathbb{Z}}(\bar{0}) = \mathbb{Z}/p\mathbb{Z},$$

and for every  $\bar{a} \neq \bar{0}$ ,

$$\text{Ann}_{\mathbb{Z}/p\mathbb{Z}}(\bar{a}) = 0.$$

**Example:**  $\mathbb{Z}/5\mathbb{Z}$

The elements are

$$\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}.$$

Then:

$$\text{Ann}(\bar{0}) = \mathbb{Z}/5\mathbb{Z},$$

but

$$\text{Ann}(\bar{1}) = 0, \quad \text{Ann}(\bar{2}) = 0, \quad \text{Ann}(\bar{3}) = 0, \quad \text{Ann}(\bar{4}) = 0.$$

Indeed, for example, if  $\bar{r}\bar{2} = \bar{0}$ , then multiply by the inverse of  $\bar{2}$ , namely  $\bar{3}$ , to get  $\bar{r} = 0$ .

**Conclusion.** Fields have no nonzero zero divisors, and therefore all nonzero annihilators are trivial.

### 1.6.5 5. Annihilators in $\mathbb{Z}/n\mathbb{Z}$

Now let

$$A = \mathbb{Z}/n\mathbb{Z}.$$

This is much more interesting when  $n$  is composite.

For  $\bar{a} \in \mathbb{Z}/n\mathbb{Z}$ ,

$$\text{Ann}(\bar{a}) = \{\bar{r} \in \mathbb{Z}/n\mathbb{Z} : \bar{r}\bar{a} = \bar{0}\}.$$

This means

$$ra \equiv 0 \pmod{n},$$

or equivalently,

$$n \mid ra.$$

Let

$$d = \gcd(n, a).$$

Then one has the concrete formula

$$\text{Ann}_{\mathbb{Z}/n\mathbb{Z}}(\bar{a}) = \left(\frac{\bar{n}}{\bar{d}}\right).$$

**Why?** Write

$$n = dn_1, \quad a = da_1, \quad \gcd(n_1, a_1) = 1.$$

Then

$$n \mid ra \iff dn_1 \mid rda_1 \iff n_1 \mid ra_1.$$

Since  $\gcd(n_1, a_1) = 1$ , this is equivalent to

$$n_1 \mid r.$$

So  $r$  must be a multiple of  $n/d$ , exactly as claimed.

**Example:**  $\mathbb{Z}/6\mathbb{Z}$

**Annihilator of  $\bar{2}$ .** Here

$$\gcd(6, 2) = 2, \quad \frac{6}{2} = 3.$$

So

$$\text{Ann}(\bar{2}) = (\bar{3}) = \{\bar{0}, \bar{3}\}.$$

Indeed,

$$\bar{0} \cdot \bar{2} = \bar{0}, \quad \bar{3} \cdot \bar{2} = \bar{6} = \bar{0}.$$

**Annihilator of  $\bar{3}$ .** Here

$$\gcd(6, 3) = 3, \quad \frac{6}{3} = 2.$$

So

$$\text{Ann}(\bar{3}) = (\bar{2}) = \{\bar{0}, \bar{2}, \bar{4}\}.$$

Indeed,

$$\bar{2} \cdot \bar{3} = \bar{0}, \quad \bar{4} \cdot \bar{3} = \bar{12} = \bar{0}.$$

**Annihilator of  $\bar{4}$ .** Since  $\gcd(6, 4) = 2$ , we get

$$\text{Ann}(\bar{4}) = (\bar{3}) = \{\bar{0}, \bar{3}\}.$$

**Annihilator of  $\bar{1}$ .** Since  $\gcd(6, 1) = 1$ ,

$$\text{Ann}(\bar{1}) = (\bar{6}) = 0.$$

Thus the nonzero elements  $\bar{2}, \bar{3}, \bar{4}$  have nontrivial annihilators, while  $\bar{1}, \bar{5}$  do not.

**Example:**  $\mathbb{Z}/12\mathbb{Z}$

**Annihilator of  $\bar{3}$ .**

$$\gcd(12, 3) = 3, \quad \frac{12}{3} = 4.$$

Hence

$$\text{Ann}(\bar{3}) = (\bar{4}) = \{\bar{0}, \bar{4}, \bar{8}\}.$$

**Annihilator of  $\bar{4}$ .**

$$\gcd(12, 4) = 4, \quad \frac{12}{4} = 3.$$

Hence

$$\text{Ann}(\bar{4}) = (\bar{3}) = \{\bar{0}, \bar{3}, \bar{6}, \bar{9}\}.$$

**Annihilator of  $\bar{6}$ .**

$$\gcd(12, 6) = 6, \quad \frac{12}{6} = 2.$$

Hence

$$\text{Ann}(\bar{6}) = (\bar{2}) = \{\bar{0}, \bar{2}, \bar{4}, \bar{6}, \bar{8}, \bar{10}\}.$$

So in  $\mathbb{Z}/n\mathbb{Z}$ , annihilators can be computed explicitly using gcd.

**Deep point.** In  $\mathbb{Z}/n\mathbb{Z}$ ,

$$\bar{a} \text{ is a zero divisor} \iff \gcd(n, a) \neq 1 \iff \text{Ann}(\bar{a}) \neq 0.$$



### 1.6.6 6. Annihilators in polynomial rings $k[x]$

Let  $k$  be a field. Since  $k[x]$  is a domain, every nonzero polynomial has trivial annihilator:

$$\text{Ann}_{k[x]}(f) = 0 \quad \text{for all } f \neq 0.$$

#### Examples

$$\text{Ann}(x) = 0, \quad \text{Ann}(x^2) = 0, \quad \text{Ann}(x^2 + 1) = 0, \quad \text{Ann}((x-1)(x+2)) = 0.$$

So in ordinary polynomial rings over a field, annihilators do not become interesting until we pass to quotients.

### 1.6.7 7. Annihilators in $k[x]/(x^2)$

Let

$$A = k[x]/(x^2).$$

Then  $\bar{x} \neq 0$ , but

$$\bar{x}^2 = 0.$$

So  $\bar{x}$  has a nontrivial annihilator.

**Annihilator of  $\bar{x}$ .** We compute

$$\text{Ann}_A(\bar{x}) = \{\bar{f} : \bar{f}\bar{x} = 0\}.$$

This means

$$fx \in (x^2).$$

So  $f$  must be divisible by  $x$ . Therefore

$$\text{Ann}_A(\bar{x}) = (\bar{x}).$$

**General element.** Every element of  $A$  has the form

$$a + b\bar{x}, \quad a, b \in k.$$

Now:

- if  $a \neq 0$ , then  $a + b\bar{x}$  is a unit, so

$$\text{Ann}(a + b\bar{x}) = 0;$$

- if  $a = 0$  and  $b \neq 0$ , then  $a + b\bar{x} = b\bar{x}$ , and

$$\text{Ann}(b\bar{x}) = (\bar{x}).$$

So the annihilators in this ring are:

$$\text{Ann}(0) = A, \quad \text{Ann}(\bar{x}) = (\bar{x}), \quad \text{Ann}(1) = 0, \quad \text{Ann}(1 + \bar{x}) = 0.$$

**Interpretation.** This ring has only one nonzero prime ideal, namely  $(\bar{x})$ , and  $\bar{x}$  is exactly the nilpotent element killed by that prime.

**1.6.8 8. Annihilators in  $k[x]/(x(x-1))$** 

Let

$$A = k[x]/(x(x-1)).$$

Then

$$\bar{x}(\bar{x} - 1) = 0.$$

**Annihilator of  $\bar{x}$ .** Since  $(\bar{x} - 1)\bar{x} = 0$ , we have

$$(\bar{x} - 1) \subseteq \text{Ann}(\bar{x}).$$

Conversely, if  $\bar{f}\bar{x} = 0$ , then

$$fx \in (x(x-1)).$$

So  $fx$  is divisible by  $x(x-1)$ , hence  $f$  is divisible by  $x-1$ . Therefore

$$\text{Ann}(\bar{x}) = (\bar{x} - 1).$$

**Annihilator of  $\bar{x} - 1$ .** Similarly,

$$\text{Ann}(\bar{x} - 1) = (\bar{x}).$$

Thus:

$$\text{Ann}(\bar{x}) = (\bar{x} - 1), \quad \text{Ann}(\bar{x} - 1) = (\bar{x}).$$

**Interpretation.** Each component kills the other one. This is the algebraic reflection of the decomposition into the two points  $x = 0$  and  $x = 1$ .

**1.6.9 9. Annihilators in  $k[x, y]/(xy)$** 

Let

$$A = k[x, y]/(xy).$$

Then

$$\bar{x}\bar{y} = 0.$$

**Annihilator of  $\bar{x}$ .** We want

$$\bar{f}\bar{x} = 0 \iff fx \in (xy).$$

This means  $fx$  is divisible by  $xy$ , so  $f$  must be divisible by  $y$ . Hence

$$\text{Ann}(\bar{x}) = (\bar{y}).$$

**Annihilator of  $\bar{y}$ .** Similarly,

$$\text{Ann}(\bar{y}) = (\bar{x}).$$

**Annihilator of  $\bar{x} + \bar{y}$ .** Now ask whether  $\bar{x} + \bar{y}$  has a nontrivial annihilator. Suppose

$$\bar{f}(\bar{x} + \bar{y}) = 0.$$

Then

$$f(x + y) \in (xy).$$

But  $x + y$  is not divisible by  $x$  or by  $y$ , so the only way this product lies in  $(xy)$  is that  $f \in (xy)$ , i.e.  $\bar{f} = 0$ . Therefore

$$\text{Ann}(\bar{x} + \bar{y}) = 0.$$

So in this ring:

$$\text{Ann}(\bar{x}) = (\bar{y}), \quad \text{Ann}(\bar{y}) = (\bar{x}), \quad \text{Ann}(\bar{x} + \bar{y}) = 0.$$

**Concrete lesson.** Not every nonunit-looking element is a zero divisor. The elements  $\bar{x}$  and  $\bar{y}$  are killed by nonzero ideals, but  $\bar{x} + \bar{y}$  is not.

### 1.6.10 10. Annihilators in $k[x, y]/(x^2, xy)$

Let

$$A = k[x, y]/(x^2, xy).$$

Then

$$\bar{x}^2 = 0, \quad \bar{x}\bar{y} = 0.$$

**Annihilator of  $\bar{x}$ .** Since both  $\bar{x}$  and  $\bar{y}$  kill  $\bar{x}$ , we have

$$(\bar{x}, \bar{y}) \subseteq \text{Ann}(\bar{x}).$$

Conversely, if  $\bar{f}\bar{x} = 0$ , then

$$fx \in (x^2, xy) = x(x, y).$$

So  $f \in (x, y)$ . Therefore

$$\text{Ann}(\bar{x}) = (\bar{x}, \bar{y}).$$

**Annihilator of  $\bar{y}$ .** Suppose  $\bar{f}\bar{y} = 0$ . Then

$$fy \in (x^2, xy).$$

Since  $x^2$  and  $xy$  are both divisible by  $x$ , and  $fy$  contains no forced factor  $x$ , the only possibility is that  $f \in (x)$  and the remaining factor still vanishes. In fact one checks that no nonzero class kills  $\bar{y}$ , so

$$\text{Ann}(\bar{y}) = 0.$$

**Annihilator of  $\bar{x} + \bar{y}$ .** Again,  $\bar{y}$  cannot be killed nontrivially, so  $\bar{x} + \bar{y}$  is not killed by all of  $(\bar{x}, \bar{y})$ . In fact

$$\text{Ann}(\bar{x} + \bar{y}) = 0.$$

Thus:

$$\text{Ann}(\bar{x}) = (\bar{x}, \bar{y}), \quad \text{Ann}(\bar{y}) = 0, \quad \text{Ann}(\bar{x} + \bar{y}) = 0.$$

**Interpretation.** The element  $\bar{x}$  is very fragile: it is killed by the maximal ideal  $(\bar{x}, \bar{y})$ . But  $\bar{y}$  behaves much more like an element in a domain.

### 1.6.11 11. Annihilators and associated primes

A prime ideal  $\mathfrak{p} \subseteq A$  is associated if

$$\mathfrak{p} = \text{Ann}_A(a)$$

for some  $a \in A$ .

So annihilators are the direct source of associated primes.

#### Examples

**In  $k[x]/(x^2)$ :**

$$\text{Ann}(\bar{x}) = (\bar{x}),$$

so

$$\text{Ass}\left(k[x]/(x^2)\right) = \{(\bar{x})\}.$$

**In  $k[x]/(x(x-1))$ :**

$$\text{Ann}(\bar{x}) = (\bar{x} - 1), \quad \text{Ann}(\bar{x} - 1) = (\bar{x}),$$

hence

$$\text{Ass}\left(k[x]/(x(x-1))\right) = \{(\bar{x}), (\bar{x} - 1)\}.$$

**In  $k[x, y]/(xy)$ :**

$$\text{Ann}(\bar{x}) = (\bar{y}), \quad \text{Ann}(\bar{y}) = (\bar{x}),$$

hence

$$\text{Ass}\left(k[x, y]/(xy)\right) = \{(\bar{x}), (\bar{y})\}.$$

**In  $k[x, y]/(x^2, xy)$ :**

$$\text{Ann}(\bar{x}) = (\bar{x}, \bar{y}),$$

so  $(\bar{x}, \bar{y})$  is associated. Also the minimal prime  $(\bar{x})$  is associated in this Noetherian example. Therefore

$$\text{Ass}\left(k[x, y]/(x^2, xy)\right) = \{(\bar{x}), (\bar{x}, \bar{y})\}.$$

### 1.6.12 12. Annihilators and minimal primes

Minimal primes describe the main geometric components. Associated primes come from annihilators and may include embedded primes.

#### Comparison table

| $A$                      | sample element $a$ | $\text{Ann}_A(a)$    | relevant primes                 |
|--------------------------|--------------------|----------------------|---------------------------------|
| $\mathbb{Z}/5\mathbb{Z}$ | $\bar{2}$          | $0$                  | no nonzero annihilator          |
| $\mathbb{Z}/6\mathbb{Z}$ | $\bar{2}$          | $(\bar{3})$          | $(\bar{3})$                     |
| $k[x]/(x^2)$             | $\bar{x}$          | $(\bar{x})$          | $(\bar{x})$                     |
| $k[x]/(x(x-1))$          | $\bar{x}$          | $(\bar{x}-1)$        | $(\bar{x}), (\bar{x}-1)$        |
| $k[x, y]/(xy)$           | $\bar{x}$          | $(\bar{y})$          | $(\bar{x}), (\bar{y})$          |
| $k[x, y]/(x^2, xy)$      | $\bar{x}$          | $(\bar{x}, \bar{y})$ | $(\bar{x}), (\bar{x}, \bar{y})$ |

### 1.6.13 13. Final summary

The annihilator is one of the most concrete and useful constructions in commutative algebra:

$$\text{Ann}_A(a) = \{r \in A : ra = 0\}.$$

Its behavior is easy to understand in important examples:

- in a field such as  $\mathbb{Z}/p\mathbb{Z}$ , every nonzero annihilator is trivial;
- in  $\mathbb{Z}/n\mathbb{Z}$ , annihilators are computed by gcd:

$$\text{Ann}(\bar{a}) = \left( \frac{\overline{n}}{\gcd(n, a)} \right);$$

- in a domain such as  $k[x]$ , every nonzero annihilator is trivial;
- in quotient rings such as  $k[x]/(x^2)$  or  $k[x, y]/(xy)$ , annihilators become nontrivial and reveal zero-divisor structure;
- prime annihilators are exactly the source of associated primes.

Thus annihilators form the bridge between concrete multiplication computations and the abstract prime-ideal structure of the ring.

## 1.7 Annihilators, minimal primes, and associated primes for modules: concrete examples

### 1.7.1 1. Annihilators for modules: the basic definitions

Let  $R$  be a ring and let  $M$  be an  $R$ -module.

For an element  $m \in M$ , the annihilator of  $m$  is

$$\text{Ann}_R(m) = \{r \in R : rm = 0\}.$$

This is an ideal of  $R$ .

The annihilator of the whole module  $M$  is

$$\text{Ann}_R(M) = \{r \in R : rM = 0\}.$$

Equivalently,

$$\text{Ann}_R(M) = \bigcap_{m \in M} \text{Ann}_R(m).$$

**Interpretation.**

- $\text{Ann}_R(m)$  tells us which scalars kill one specific element  $m$ .
- $\text{Ann}_R(M)$  tells us which scalars kill the entire module.

These two ideals are usually different.

## 1.7.2 2. Associated primes and minimal primes of a module

A prime ideal  $\mathfrak{p} \subseteq R$  is an *associated prime* of the  $R$ -module  $M$  if

$$\mathfrak{p} = \text{Ann}_R(m)$$

for some  $m \in M$ .

We write

$$\text{Ass}_R(M)$$

for the set of associated primes of  $M$ .

If  $M$  is finitely generated over a Noetherian ring, one defines the minimal primes of  $M$  as the minimal primes in the support of  $M$ , equivalently the minimal primes over  $\text{Ann}_R(M)$ . We write

$$\text{Min}_R(M)$$

for this set.

**Interpretation.**

- $\text{Min}_R(M)$  describes the main geometric components seen by the module.
- $\text{Ass}_R(M)$  records prime annihilators of actual elements of the module.

In the Noetherian finitely generated setting,

$$\text{Min}_R(M) \subseteq \text{Ass}_R(M),$$

and the inclusion may be strict.

### 1.7.3 3. The cyclic module $R/I$

A very important case is the cyclic module

$$M = R/I.$$

Then the element  $\bar{1} = 1 + I \in R/I$  satisfies

$$\text{Ann}_R(\bar{1}) = I.$$

More generally, for  $\bar{f} = f + I \in R/I$ ,

$$\text{Ann}_R(\bar{f}) = \{r \in R : rf \in I\} = (I : f),$$

the colon ideal.

So cyclic modules give a very concrete way to compute annihilators.

### 1.7.4 4. First examples over $\mathbb{Z}$

**Example 1:**  $M = \mathbb{Z}$  as a  $\mathbb{Z}$ -module

Take  $R = \mathbb{Z}$  and  $M = \mathbb{Z}$ .

For any nonzero integer  $m$ ,

$$\text{Ann}_{\mathbb{Z}}(m) = 0,$$

because if  $rm = 0$  in  $\mathbb{Z}$ , then  $r = 0$ .

Thus

$$\text{Ann}_{\mathbb{Z}}(\mathbb{Z}) = 0.$$

Since  $(0)$  is prime in  $\mathbb{Z}$ , we get

$$\text{Ass}_{\mathbb{Z}}(\mathbb{Z}) = \{(0)\}, \quad \text{Min}_{\mathbb{Z}}(\mathbb{Z}) = \{(0)\}.$$

So the free module  $\mathbb{Z}$  behaves like a torsion-free object.

**Example 2:**  $M = \mathbb{Z}/12\mathbb{Z}$

Now let

$$M = \mathbb{Z}/12\mathbb{Z}.$$

As a  $\mathbb{Z}$ -module, this is a cyclic module:

$$M = \mathbb{Z}/(12).$$

First,

$$\text{Ann}_{\mathbb{Z}}(M) = (12),$$

because 12 kills every class modulo 12, and no smaller ideal does.

Now compute annihilators of specific elements.

**Element  $\bar{1}$ .**

$$\text{Ann}(\bar{1}) = (12).$$

**Element  $\bar{2}$ .** We want all  $r \in \mathbb{Z}$  such that

$$r\bar{2} = \bar{0} \text{ in } \mathbb{Z}/12\mathbb{Z}.$$

This means

$$12 \mid 2r,$$

so  $r$  must be divisible by 6. Hence

$$\text{Ann}(\bar{2}) = (6).$$

**Element  $\bar{3}$ .** We need

$$12 \mid 3r,$$

so  $r$  must be divisible by 4. Thus

$$\text{Ann}(\bar{3}) = (4).$$

**Element  $\bar{4}$ .** We need

$$12 \mid 4r,$$

so  $r$  must be divisible by 3. Hence

$$\text{Ann}(\bar{4}) = (3).$$

**Element  $\bar{6}$ .** We need

$$12 \mid 6r,$$

so  $r$  must be divisible by 2. Hence

$$\text{Ann}(\bar{6}) = (2).$$

So the prime annihilators that occur are

$$(2), \quad (3).$$

Therefore

$$\text{Ass}_{\mathbb{Z}}(\mathbb{Z}/12\mathbb{Z}) = \{(2), (3)\}.$$

Also the minimal primes over  $(12)$  are exactly

$$(2), \quad (3),$$

so

$$\text{Min}_{\mathbb{Z}}(\mathbb{Z}/12\mathbb{Z}) = \{(2), (3)\}.$$

**General formula.** For  $M = \mathbb{Z}/n\mathbb{Z}$  and  $\bar{a} \in M$ ,

$$\text{Ann}_{\mathbb{Z}}(\bar{a}) = \left( \frac{n}{\gcd(n, a)} \right).$$



### 1.7.5 5. A vector-space example

Let  $k$  be a field, and let  $M = k^n$  be a  $k$ -vector space.

For every nonzero  $v \in k^n$ ,

$$\text{Ann}_k(v) = 0,$$

because only  $0 \in k$  kills a nonzero vector.

Thus

$$\text{Ann}_k(k^n) = 0, \quad \text{Ass}_k(k^n) = \{(0)\}, \quad \text{Min}_k(k^n) = \{(0)\}.$$

So over a field, modules behave very simply: nonzero elements have trivial annihilator.

### 1.7.6 6. Polynomial modules: $M = k[x]/(x^2)$

Let

$$R = k[x], \quad M = R/(x^2).$$

Then  $M$  is an  $R$ -module.

Since  $M = R/I$  with  $I = (x^2)$ ,

$$\text{Ann}_R(M) = (x^2).$$

Now compute annihilators of concrete elements.

**Element  $\bar{1}$ .**

$$\text{Ann}_R(\bar{1}) = (x^2).$$

**Element  $\bar{x}$ .** We need all  $f \in k[x]$  such that

$$f\bar{x} = 0 \text{ in } R/(x^2).$$

This means

$$fx \in (x^2),$$

so  $f$  must be divisible by  $x$ . Hence

$$\text{Ann}_R(\bar{x}) = (x).$$

**Element  $\overline{1+x}$ .** Suppose

$$f(\overline{1+x}) = 0.$$

Then

$$f(1+x) \in (x^2).$$

But  $1+x$  is not divisible by  $x$ , so the only way this can happen is  $f \in (x^2)$ . Therefore

$$\text{Ann}_R(\overline{1+x}) = (x^2).$$

Thus the prime annihilator that appears is

$$(x).$$

So

$$\text{Ass}_R(k[x]/(x^2)) = \{(x)\}.$$

Also the only minimal prime over  $(x^2)$  is

$$(x),$$

hence

$$\text{Min}_R(k[x]/(x^2)) = \{(x)\}.$$

### 1.7.7 7. Polynomial modules: $M = k[x]/(x(x-1))$

Let

$$R = k[x], \quad M = R/(x(x-1)).$$

Then

$$\text{Ann}_R(M) = (x(x-1)).$$

Now compute annihilators of elements.

**Element  $\bar{1}$ .**

$$\text{Ann}(\bar{1}) = (x(x-1)).$$

**Element  $\bar{x}$ .** We need

$$f\bar{x} = 0 \iff fx \in (x(x-1)).$$

So  $fx$  must be divisible by  $x(x-1)$ , hence  $f$  must be divisible by  $x-1$ . Therefore

$$\text{Ann}(\bar{x}) = (x-1).$$

**Element  $\overline{x-1}$ .** Similarly,

$$\text{Ann}(\overline{x-1}) = (x).$$

Thus the associated primes are

$$\text{Ass}_R(k[x]/(x(x-1))) = \{(x), (x-1)\}.$$

These are also the minimal primes over  $(x(x-1))$ , so

$$\text{Min}_R(k[x]/(x(x-1))) = \{(x), (x-1)\}.$$

**Interpretation.** The module sees two components, corresponding to the two points  $x = 0$  and  $x = 1$ .

**1.7.8 8. Polynomial modules:**  $M = k[x, y]/(xy)$ 

Let

$$R = k[x, y], \quad M = R/(xy).$$

Then

$$\text{Ann}_R(M) = (xy).$$

Now compute element annihilators.

**Element  $\bar{x}$ .** We need

$$f\bar{x} = 0 \iff fx \in (xy).$$

Thus  $f$  must be divisible by  $y$ , so

$$\text{Ann}(\bar{x}) = (y).$$

**Element  $\bar{y}$ .** Similarly,

$$\text{Ann}(\bar{y}) = (x).$$

**Element  $\overline{x+y}$ .** Suppose

$$f(\overline{x+y}) = 0.$$

Then

$$f(x+y) \in (xy).$$

Since  $x+y$  is not divisible by  $x$  or by  $y$ , the only possibility is

$$f \in (xy).$$

Therefore

$$\text{Ann}(\overline{x+y}) = (xy).$$

So the prime annihilators that occur are

$$(x), \quad (y).$$

Hence

$$\text{Ass}_R(k[x, y]/(xy)) = \{(x), (y)\}.$$

These are also the minimal primes over  $(xy)$ , so

$$\text{Min}_R(k[x, y]/(xy)) = \{(x), (y)\}.$$

**Interpretation.** The element  $\bar{x}$  is killed by the  $y$ -axis, and  $\bar{y}$  is killed by the  $x$ -axis.

**1.7.9 9. An embedded-prime module:**  $M = k[x, y]/(x^2, xy)$ 

Let

$$R = k[x, y], \quad M = R/(x^2, xy).$$

Then

$$\text{Ann}_R(M) = (x^2, xy).$$

Now compute annihilators of elements.

**Element  $\bar{1}$ .**

$$\text{Ann}(\bar{1}) = (x^2, xy).$$

**Element  $\bar{x}$ .** We need

$$f\bar{x} = 0 \iff fx \in (x^2, xy) = x(x, y).$$

This is equivalent to  $f \in (x, y)$ . Hence

$$\text{Ann}(\bar{x}) = (x, y).$$

**Element  $\bar{y}$ .** We need

$$f\bar{y} = 0 \iff fy \in (x^2, xy).$$

Since  $fy$  belongs to  $(x^2, xy) = x(x, y)$ , it must be divisible by  $x$ . Because  $x$  and  $y$  are independent variables, this forces  $f \in (x)$ . Conversely, any multiple of  $x$  kills  $\bar{y}$ , because  $xy \in (x^2, xy)$ . Therefore

$$\text{Ann}(\bar{y}) = (x).$$

Thus we get two prime annihilators:

$$(x), \quad (x, y).$$

Hence

$$\text{Ass}_R(k[x, y]/(x^2, xy)) = \{(x), (x, y)\}.$$

Now

$$\sqrt{(x^2, xy)} = (x),$$

so the only minimal prime over  $\text{Ann}_R(M)$  is

$$(x).$$

Thus

$$\text{Min}_R(k[x, y]/(x^2, xy)) = \{(x)\}.$$

Therefore

$$\text{Min}_R(M) = \{(x)\} \subsetneq \text{Ass}_R(M) = \{(x), (x, y)\}.$$

The prime  $(x, y)$  is an embedded associated prime.

**1.7.10 10. Direct sums of modules**

Let

$$R = k[x, y], \quad M = R/(x) \oplus R/(y).$$

Then

$$\text{Ann}_R(M) = \text{Ann}_R(R/(x)) \cap \text{Ann}_R(R/(y)) = (x) \cap (y) = (xy).$$

Now consider the two elements

$$m_1 = (\bar{1}, 0), \quad m_2 = (0, \bar{1}).$$

Their annihilators are:

$$\text{Ann}_R(m_1) = (x), \quad \text{Ann}_R(m_2) = (y).$$

So

$$\text{Ass}_R(M) = \{(x), (y)\}.$$

And the minimal primes over  $(xy)$  are again

$$(x), \quad (y).$$

Hence

$$\text{Min}_R(M) = \{(x), (y)\}.$$

This is a very concrete example showing that associated primes of a direct sum are the union of the associated primes of the summands.

**1.7.11 11. A free polynomial module**

Let

$$R = k[x], \quad M = R^2.$$

Then no nonzero polynomial kills all of  $R^2$ , so

$$\text{Ann}_R(M) = 0.$$

Also, for a nonzero vector such as

$$m = (1, x),$$

if  $f(1, x) = (0, 0)$ , then  $f = 0$ . Hence

$$\text{Ann}_R(m) = 0.$$

Since  $R = k[x]$  is a domain,  $(0)$  is prime. Therefore

$$\text{Ass}_R(R^2) = \{(0)\}, \quad \text{Min}_R(R^2) = \{(0)\}.$$

So free modules over domains behave like torsion-free modules.

### 1.7.12 12. Summary table

| $R$          | $M$                       | $\text{Ann}_R(M)$ | $\text{Ass}_R(M)$ | $\text{Min}_R(M)$ |
|--------------|---------------------------|-------------------|-------------------|-------------------|
| $\mathbb{Z}$ | $\mathbb{Z}$              | $(0)$             | $\{(0)\}$         | $\{(0)\}$         |
| $\mathbb{Z}$ | $\mathbb{Z}/12\mathbb{Z}$ | $(12)$            | $\{(2), (3)\}$    | $\{(2), (3)\}$    |
| $k[x]$       | $k[x]/(x^2)$              | $(x^2)$           | $\{(x)\}$         | $\{(x)\}$         |
| $k[x]$       | $k[x]/(x(x-1))$           | $(x(x-1))$        | $\{(x), (x-1)\}$  | $\{(x), (x-1)\}$  |
| $k[x, y]$    | $k[x, y]/(xy)$            | $(xy)$            | $\{(x), (y)\}$    | $\{(x), (y)\}$    |
| $k[x, y]$    | $k[x, y]/(x^2, xy)$       | $(x^2, xy)$       | $\{(x), (x, y)\}$ | $\{(x)\}$         |
| $k[x, y]$    | $R/(x) \oplus R/(y)$      | $(xy)$            | $\{(x), (y)\}$    | $\{(x), (y)\}$    |
| $k[x]$       | $R^2$                     | $(0)$             | $\{(0)\}$         | $\{(0)\}$         |

### 1.7.13 13. Final interpretation

For modules, one must always distinguish:

$$\text{Ann}_R(m) \quad \text{for one element } m \in M,$$

from

$$\text{Ann}_R(M) \quad \text{for the whole module.}$$

The annihilator of the whole module need not be prime and usually describes the global defining equations of the module. The annihilator of a specific element may be much smaller, and when it is prime, it contributes to the associated primes.

Thus:

$$\text{minimal primes of } M = \text{main geometric support of } M,$$

while

$$\text{associated primes of } M = \text{prime annihilators of actual elements of } M.$$

That is why associated primes are finer than minimal primes: they can detect embedded structure that the global annihilator  $\text{Ann}_R(M)$  does not show by itself.

## 1.8 Concrete residue classes in $\mathbb{Z}/n\mathbb{Z}$ , $\mathbb{Z}/p\mathbb{Z}$ , $k[x]/(x)$ , $A[t]/(t)$ , and $k[x, y]/(xy)$

### 1.8.1 1. General principle: what is a residue class in $R/I$ ?

Let  $R$  be a ring and  $I \subseteq R$  an ideal. An element of the quotient ring

$$R/I$$

is a residue class

$$\bar{f} = f + I = \{f + h : h \in I\}.$$

Thus  $\bar{f}$  is not just one element  $f$ , but the whole set of elements differing from  $f$  by something in  $I$ .

Two elements  $f, g \in R$  define the same class if and only if

$$\bar{f} = \bar{g} \iff f - g \in I.$$

So the quotient  $R/I$  identifies all elements that differ by an element of  $I$ .

### 1.8.2 2. Residue classes in $\mathbb{Z}/n\mathbb{Z}$

The quotient

$$\mathbb{Z}/n\mathbb{Z}$$

consists of congruence classes modulo  $n$ . For an integer  $a$ ,

$$\bar{a} = a + n\mathbb{Z} = \{a + nk : k \in \mathbb{Z}\}.$$

So  $\bar{a}$  is the set of all integers congruent to  $a$  modulo  $n$ .

**Example:**  $\mathbb{Z}/5\mathbb{Z}$

We have

$$\bar{2} = 2 + 5\mathbb{Z} = \{\dots, -8, -3, 2, 7, 12, 17, \dots\}.$$

Also

$$\bar{2} = \bar{7} = \overline{-3},$$

because

$$7 - 2 = 5, \quad -3 - 2 = -5,$$

and both are divisible by 5.

Thus the elements of  $\mathbb{Z}/5\mathbb{Z}$  are

$$\bar{0}, \bar{1}, \bar{2}, \bar{3}, \bar{4}.$$

**Example:**  $\mathbb{Z}/6\mathbb{Z}$

Similarly,

$$\bar{4} = 4 + 6\mathbb{Z} = \{\dots, -8, -2, 4, 10, 16, \dots\}.$$

Also

$$\bar{4} = \bar{10} = \overline{-2}.$$

Thus in  $\mathbb{Z}/n\mathbb{Z}$ , a residue class is simply a set of integers having the same remainder modulo  $n$ .

### 1.8.3 3. Residue classes in $\mathbb{Z}/p\mathbb{Z}$

If  $p$  is prime, then

$$\mathbb{Z}/p\mathbb{Z}$$

is a field. The residue classes are defined exactly as above:

$$\bar{a} = a + p\mathbb{Z}.$$

**Example:**  $\mathbb{Z}/7\mathbb{Z}$

$$\bar{3} = 3 + 7\mathbb{Z} = \{\dots, -11, -4, 3, 10, 17, 24, \dots\}.$$

So the only difference between  $\mathbb{Z}/p\mathbb{Z}$  and  $\mathbb{Z}/n\mathbb{Z}$  is not how classes are formed, but the algebraic structure afterward: when  $p$  is prime, every nonzero class has a multiplicative inverse.

### 1.8.4 4. Residue classes in $k[x]/(x)$

Now let  $k$  be a field. Consider

$$k[x]/(x).$$

An element is a class

$$\bar{f} = f(x) + (x).$$

Two polynomials  $f(x)$  and  $g(x)$  lie in the same class if and only if

$$f(x) - g(x) \in (x),$$

that is, if  $f(x) - g(x)$  is divisible by  $x$ .

This means precisely that  $f$  and  $g$  have the same constant term.

Indeed, if

$$f(x) = a_0 + a_1x + \dots + a_nx^n,$$

then

$$f(x) - a_0 = x(a_1 + a_2x + \dots + a_nx^{n-1}) \in (x),$$

so

$$\bar{f} = \overline{a_0}.$$

Thus every class is determined by the constant term alone.

**Class of 0**

$$\bar{0} = (x) = \{xq(x) : q(x) \in k[x]\}.$$

So this class consists of all polynomials divisible by  $x$ :

$$x, \quad x^2, \quad 3x + 7x^2, \quad x(x^3 + 1), \dots$$



**Class of 1**

$$\bar{1} = 1 + (x) = \{1 + xq(x) : q(x) \in k[x]\}.$$

Examples:

$$1, \quad 1 + x, \quad 1 + 3x^2, \quad 1 + x + x^4, \dots$$

**Class of  $a \in k$** 

$$\bar{a} = a + (x) = \{a + xq(x) : q(x) \in k[x]\}.$$

Thus in  $k[x]/(x)$ , two polynomials are equal if and only if they have the same constant term. Therefore

$$k[x]/(x) \cong k.$$

**1.8.5 5. Residue classes in  $A[t]/(t)$** 

Let  $A$  be any commutative ring. Consider

$$A[t]/(t).$$

An element is a class

$$\bar{f} = f(t) + (t).$$

If

$$f(t) = a_0 + a_1t + \dots + a_nt^n,$$

then

$$f(t) - a_0 \in (t),$$

so

$$\bar{f} = \overline{a_0}.$$

Thus every class is again determined by the constant term.

**Class of  $a \in A$** 

$$\bar{a} = a + (t) = \{a + tq(t) : q(t) \in A[t]\}.$$

**Class of 0**

$$\bar{0} = (t) = \{tq(t) : q(t) \in A[t]\}.$$

So  $A[t]/(t)$  behaves exactly like  $A$ , and in fact

$$A[t]/(t) \cong A.$$

**Example**

In  $\mathbb{Z}[t]/(t)$ ,

$$\overline{7 + 3t + t^2} = \bar{7},$$

and the class  $\bar{7}$  consists of all polynomials

$$7 + tq(t).$$

**1.8.6 6. Residue classes in  $k[x, y]/(x)$** 

Before treating  $k[x, y]/(xy)$ , it is useful to look at

$$k[x, y]/(x).$$

Here two polynomials are equivalent if and only if their difference is divisible by  $x$ .

So every polynomial becomes equal to what remains after setting  $x = 0$ . In other words, only the  $y$ -part survives.

If

$$f(x, y) = g(y) + xh(x, y),$$

then

$$\bar{f} = \overline{g(y)}.$$

Thus every class has a representative depending only on  $y$ , and

$$k[x, y]/(x) \cong k[y].$$

**Example**

$$x^2 + xy + y^3 + 2y + 5$$

has class

$$y^3 + 2y + 5.$$

So the class of  $g(y)$  is

$$g(y) + (x) = \{g(y) + xh(x, y) : h(x, y) \in k[x, y]\}.$$

**1.8.7 7. Residue classes in  $k[x, y]/(xy)$** 

Now consider

$$B = k[x, y]/(xy).$$

An element is a class

$$\bar{f} = f(x, y) + (xy).$$

Two polynomials are in the same class if and only if their difference is divisible by  $xy$ :

$$\bar{f} = \bar{g} \iff f - g \in (xy).$$

This means that all mixed terms containing both  $x$  and  $y$  vanish in the quotient.

**What survives modulo  $(xy)$ ?**

Write

$$f(x, y) = \sum_{i, j \geq 0} a_{ij} x^i y^j.$$

If  $i \geq 1$  and  $j \geq 1$ , then  $x^i y^j$  is divisible by  $xy$ , so its class is zero in the quotient.

Therefore the only terms that survive are:

- the constant term,
- pure powers of  $x$ :  $x, x^2, x^3, \dots$ ,
- pure powers of  $y$ :  $y, y^2, y^3, \dots$

So every class has a representative of the form

$$a + p(x) + q(y),$$

where  $a \in k$ ,  $p(x) \in xk[x]$ , and  $q(y) \in yk[y]$ .

Equivalently, the whole class is

$$a + p(x) + q(y) + (xy),$$

that is, all polynomials of the form

$$a + p(x) + q(y) + xy h(x, y)$$

with  $h(x, y) \in k[x, y]$ .

**Class of 0**

$$\bar{0} = (xy) = \{xy h(x, y) : h(x, y) \in k[x, y]\}.$$

Examples of polynomials in this class are:

$$xy, \quad x^2y, \quad xy^2, \quad 3xy + 7x^4y^3, \dots$$

All of these become zero in the quotient.

**Class of  $\bar{x}$** 

$$\bar{x} = x + (xy) = \{x + xy h(x, y) : h(x, y) \in k[x, y]\}.$$

Examples:

$$x, \quad x + xy, \quad x + 7x^2y, \quad x + x^5y^2, \dots$$

All these are equal to  $\bar{x}$  in the quotient.

However,

$$x + x^2$$

is *not* in the same class as  $x$ , because

$$(x + x^2) - x = x^2,$$

and  $x^2 \notin (xy)$ .

So

$$\overline{x + x^2} \neq \bar{x}.$$

### Class of $\bar{y}$

Similarly,

$$\bar{y} = y + (xy) = \{y + xy h(x, y) : h(x, y) \in k[x, y]\}.$$

Examples:

$$y, \quad y + xy, \quad y + 5xy^2, \quad y + x^3y, \dots$$

### Class of $\overline{x + y^2 + 3}$

This class is

$$x + y^2 + 3 + (xy).$$

So it consists of all polynomials of the form

$$x + y^2 + 3 + xy h(x, y).$$

Examples in this class:

$$\begin{aligned} & x + y^2 + 3, \\ & x + y^2 + 3 + xy, \\ & x + y^2 + 3 + 7x^2y^5, \\ & x + y^2 + 3 + xy(x + y + 1). \end{aligned}$$

### Class of $\overline{x^2 + y^3}$

This class is

$$x^2 + y^3 + (xy).$$

So it contains:

$$x^2 + y^3, \quad x^2 + y^3 + xy, \quad x^2 + y^3 + x^7y^2, \dots$$

But

$$x^2 + y^3 + x$$

is not in this class, because the difference is  $x$ , which is not divisible by  $xy$ .

### 1.8.8 8. What are $\bar{x}$ and $\bar{y}$ concretely in $k[x, y]/(xy)$ ?

They are the residue classes

$$\bar{x} = x + (xy), \quad \bar{y} = y + (xy).$$

So:

- $\bar{x}$  is the class of all polynomials congruent to  $x$  modulo  $(xy)$ ,
- $\bar{y}$  is the class of all polynomials congruent to  $y$  modulo  $(xy)$ .

They are not zero, because

$$x \notin (xy), \quad y \notin (xy).$$

But their product is zero:

$$\bar{x}\bar{y} = \overline{xy} = 0.$$

Thus  $\bar{x}$  and  $\bar{y}$  are nonzero zero divisors.

Also,

$$\bar{x}^2 \neq 0, \quad \bar{y}^2 \neq 0,$$

because

$$x^2 \notin (xy), \quad y^2 \notin (xy).$$

So only mixed terms vanish, not pure powers of  $x$  or  $y$ .

### 1.8.9 9. Comparison of the main examples

The quotient always identifies elements that differ by something in the defining ideal:

- in  $\mathbb{Z}/n\mathbb{Z}$ , one adds multiples of  $n$ :

$$\bar{a} = a + n\mathbb{Z};$$

- in  $k[x]/(x)$ , one adds multiples of  $x$ :

$$\bar{a} = a + (x) = \{a + xq(x)\};$$

- in  $A[t]/(t)$ , one adds multiples of  $t$ :

$$\bar{a} = a + (t) = \{a + tq(t)\};$$

- in  $k[x, y]/(x)$ , one adds multiples of  $x$ :

$$g(y) + (x) = \{g(y) + xh(x, y)\};$$

- in  $k[x, y]/(xy)$ , one adds multiples of  $xy$ :

$$a + p(x) + q(y) + (xy) = \{a + p(x) + q(y) + xy h(x, y)\}.$$

Thus:

- in  $k[x]/(x)$ , the class is determined only by the constant term;
- in  $A[t]/(t)$ , likewise only the constant term survives;
- in  $k[x, y]/(x)$ , only the  $y$ -part survives;
- in  $k[x, y]/(xy)$ , the constant term, pure  $x$ -terms, and pure  $y$ -terms survive, while mixed  $xy$ -terms vanish.

### 1.8.10 10. Final summary

A residue class in any quotient  $R/I$  is always

$$f + I = \{f + h : h \in I\}.$$

So:

- in  $\mathbb{Z}/n\mathbb{Z}$ , two integers are equal if they differ by a multiple of  $n$ ;
- in  $k[x]/(x)$ , two polynomials are equal if they have the same constant term;
- in  $A[t]/(t)$ , two polynomials are equal if they have the same constant term;
- in  $k[x, y]/(x)$ , two polynomials are equal if they become equal after setting  $x = 0$ ;
- in  $k[x, y]/(xy)$ , two polynomials are equal if they differ by a multiple of  $xy$ , so mixed terms vanish.

In particular,

$$\bar{x} = 0 \text{ in } k[x]/(x),$$

because  $x \in (x)$ , but

$$\bar{x} \neq 0 \text{ in } k[x, y]/(xy),$$

because  $x \notin (xy)$ .